

Cable theory is a mathematical model used to calculate currents and voltages along and across axon membranes. The model treats small sections of the membrane as the resistor-capacitor circuit shown in Figure 1. An electric neuronal signal enters the circuit through the input node. A portion of the signal remains inside the axon and continues to the next section of membrane via Output 1, and the rest of the signal exits to the extracellular fluid via Output 2.

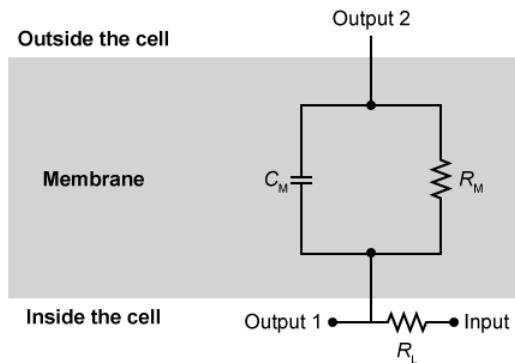


Figure 1 Circuit unit used to model axon membranes

Researchers studied cable theory by recreating the circuit in Figure 1 using a $2\ \mu\text{F}$ capacitor for C_M , a $200\ \Omega$ resistor for R_M , and a $1,000\ \Omega$ resistor for R_L . The researchers sent an electrical signal through the input and discovered that the capacitor could be modeled as a resistor whose resistance varied with the signal's frequency, as shown in Figure 2.

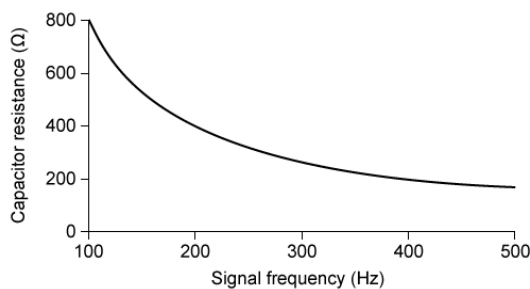


Figure 2 Capacitor equivalent resistance vs. signal frequency for C_M

To test the integrity of the capacitor, researchers maintained a voltage of 50 mV across C_M and R_M for several minutes. During this time, how much energy was stored in the capacitor?

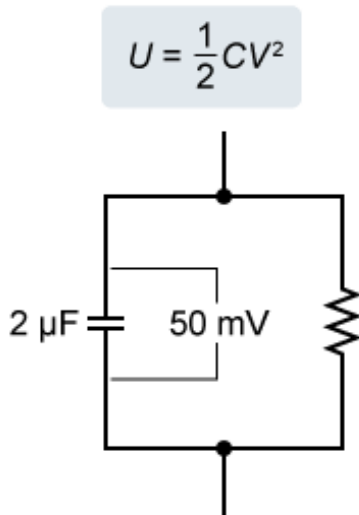
- ✓ ☒ A. $2.5 \times 10^{-9}\ \text{J}$ [43%]
☐ B. $5.0 \times 10^{-9}\ \text{J}$ [18%]
☐ C. $5.0 \times 10^{-8}\ \text{J}$ [18%]
☐ D. $1.0 \times 10^{-7}\ \text{J}$ [19%]

Correct

44% answered correctly

Explanation:

Explanation:



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Capacitance is the ability of a capacitor to store **electric charges** and is measured in **farads (F)**. The simplest capacitor is a **parallel plate capacitor**, which consists of two evenly spaced, flat metallic plates. When a voltage is applied across a parallel plate capacitor, the plate connected to the higher voltage accumulates positive charges, and the plate connected to the lower voltage accumulates an equal amount of negative charges.

The **energy** U in a capacitor is stored in the electric field created between the oppositely charged plates. Its magnitude depends on the capacitance C and the voltage V across the plates of the capacitor:

$$U = \frac{1}{2}CV^2$$

The 50 mV source is connected to the parallel circuit consisting of C_M and R_M . Because these two components are in parallel, they have the same V across them. Therefore, the V across C_M (and the value of U) only depends on the 50 mV source and is independent of R_M .

C is given as 2 μF, and V is given as 50 mV. Using $1 \mu\text{F} = 10^{-6} \text{ F}$ and $1 \text{ mV} = 10^{-3} \text{ V}$:

$$C = 2 \mu\text{F} \times \frac{10^{-6} \text{ F}}{1 \mu\text{F}} = 2 \times 10^{-6} \text{ F}$$

$$V = 50 \text{ mV} \times \frac{10^{-3} \text{ V}}{1 \text{ mV}} = 5 \times 10^{-2} \text{ V}$$

Using the above values in the capacitor energy equation:

$$U = \frac{1}{2}(2 \times 10^{-6} \text{ F})(5 \times 10^{-2} \text{ V})^2 = 2.5 \times 10^{-9} \text{ J}$$

(see detailed [calculation steps](#)). The energy stored in the capacitor is **$2.5 \times 10^{-9} \text{ J}$** .

(Choice D) The amount of charge stored in a capacitor is $Q = CV$. If this equation is incorrectly used to calculate the charge instead of the energy stored in the capacitor, the result is $1.0 \times 10^{-7} \text{ C}$.

Educational objective:

Capacitance is the ability of a capacitor to store electric charges. The amount of energy U stored in a charged capacitor depends on its capacitance C and the voltage V across its plates: $U = \frac{1}{2}CV^2$.

Time spent:0 sec

Subject:
Physics

QID:401065

System:
Electrostatics and Circuits

Cable theory is a mathematical model used to calculate currents and voltages along and across axon membranes. The model treats small sections of the membrane as the resistor-capacitor circuit shown in Figure 1. An electric neuronal signal enters the circuit through the input node. A portion of the signal remains inside the axon and continues to the next section of membrane via Output 1, and the rest of the signal exits to the extracellular fluid via Output 2.

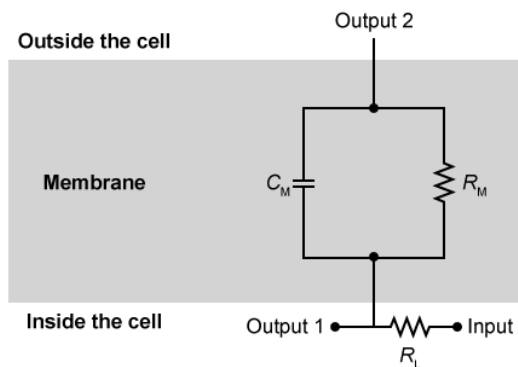


Figure 1 Circuit unit used to model axon membranes

Researchers studied cable theory by recreating the circuit in Figure 1 using a $2\ \mu\text{F}$ capacitor for C_M , a $200\ \Omega$ resistor for R_M , and a $1,000\ \Omega$ resistor for R_L . The researchers sent an electrical signal through the input and discovered that the capacitor could be modeled as a resistor whose resistance varied with the signal's frequency, as shown in Figure 2.

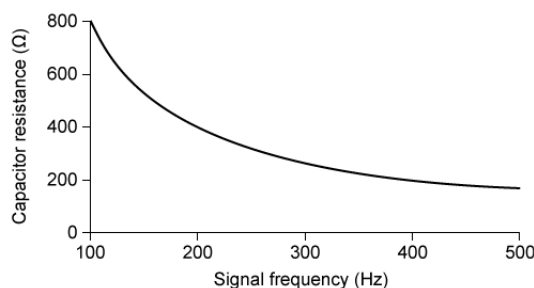


Figure 2 Capacitor equivalent resistance vs. signal frequency for C_M

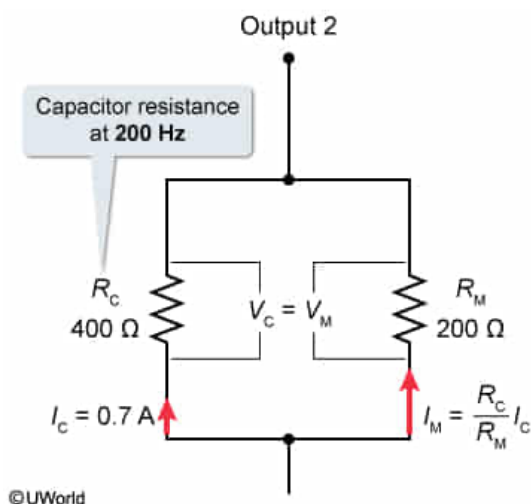
At 200 Hz, the current in C_M is 0.7 A. What is the current in R_M ?

- ☐ A. 0.35 A [12%]
- ☐ B. 0.70 A [30%]
- ☒ C. 1.4 A [51%]
- ☐ D. 2.8 A [5%]

Correct

50% answered correctly

Explanation:



According to the passage, C_M can be modeled as a resistor R_C with the resistance values shown in Figure 2. Therefore, the circuit consists of resistors R_C and R_M in parallel. Circuit components in **parallel** are connected to each other at both ends and have the **same voltage** across them.

$$V_C = V_M$$

Ohm's law states that the **voltage across a resistor** equals the product of its resistance and the current flowing through it ($V = IR$). Because R_C and R_M are connected in parallel, the product of their currents and resistances must be equal:

$$I_C R_C = I_M R_M$$

The current I_M through the resistor is obtained by dividing both sides of the equation by R_M :

$$I_M = \frac{R_C}{R_M} I_C$$

Based on Figure 2, C_M is modeled as a $400\ \Omega$ resistor when the signal frequency is 200 Hz, so $R_C = 400\ \Omega$. The current through R_C is given as $I_C = 0.7\text{ A}$, and the resistance of R_M is given as $200\ \Omega$. Therefore, the current through R_M is calculated as:

$$I_M = \left(\frac{400\ \Omega}{200\ \Omega} \right) (0.7\text{ A}) = (2)(0.7\text{ A}) = 1.4\text{ A}$$

(Choice A) Because the voltages of parallel resistors are equal, current and resistance are inversely proportional ($V = IR$). Resistors R_C and R_M have the same voltage across them, and therefore the larger of the two resistors will have the smaller current.

(Choice B) Resistors in parallel have the same current only if they have the same resistance.

Educational objective:

Circuit components in parallel have the same voltage across them. The current through a

resistor can be determined from its resistance and the voltage across it using Ohm law: $V = IR$.

Subject:
Physics

System:
Electrostatics and Circuits

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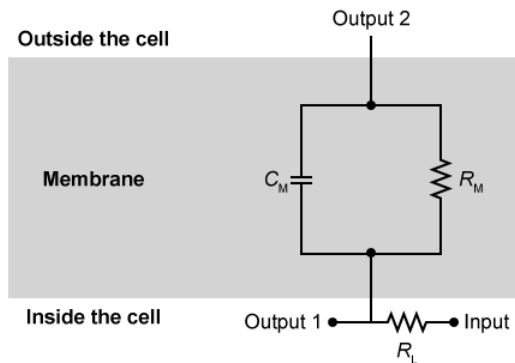


Figure 1 Circuit unit used to model axon membranes

Researchers studied cable theory by recreating the circuit in Figure 1 using a $2\ \mu\text{F}$ capacitor for C_M , a $200\ \Omega$ resistor for R_M , and a $1,000\ \Omega$ resistor for R_L . The researchers sent an electrical signal through the input and discovered that the capacitor could be modeled as a resistor whose resistance varied with the signal's frequency, as shown in Figure 2.

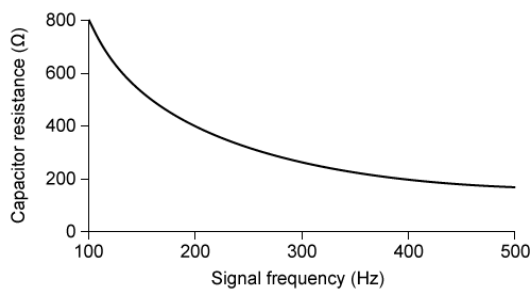


Figure 2 Capacitor equivalent resistance vs. signal frequency for C_M

If the voltage across C_M and R_M is 2 V, how much power is dissipated from R_M ?

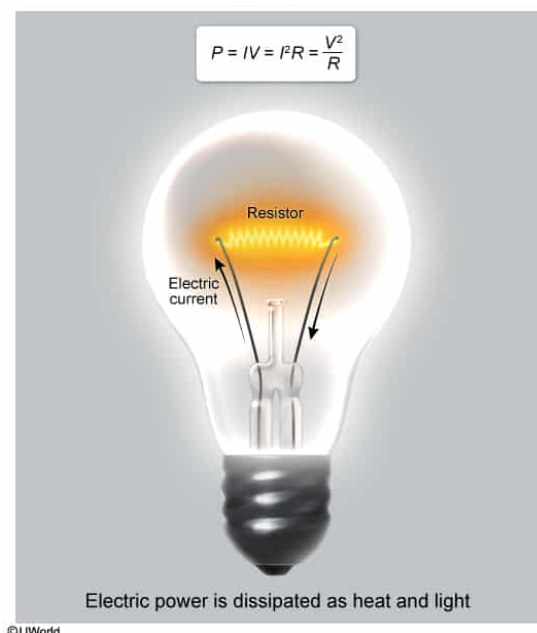
- ✓ ☒ A. 20 mW [53%]
- ☐ B. 40 mW [18%]
- ☐ C. 200 mW [23%]
- ☐ D. 800 mW [5%]

Correct

52% answered correctly

Explanation:

Electric power



Electric power is the rate of electric energy transfer, and it is **dissipated as heat** in resistors (devices that resist the flow of electric current). The amount of electric power P in an electrical component is the product its current I and voltage V across it:

$$P = IV$$

Ohm's law states that the **voltage across a resistor** is the product of its current and resistance ($V = IR$). Rearranging, the current flowing through a resistor can be calculated:

$$I = \frac{V}{R}$$

Combining these two equations, the power dissipated by a resistor can be calculated from voltage and resistance:

$$P = \left(\frac{V}{R}\right)V = \frac{V^2}{R}$$

The resistance of R_M is $200\ \Omega$ and the power dissipated by R_M is:

$$P = \frac{(2\text{ V})^2}{200\ \Omega} = \frac{4\text{ V}^2}{200\ \Omega} = 0.02 \frac{\text{V}^2}{\Omega} = 0.02\text{ W}$$

Because **1 W = 1,000 mW**, the power dissipated by R_M in mW is:

$$0.02\text{ W} \times \frac{1,000\text{ mW}}{1\text{ W}} = 20\text{ mW}$$

(Choice B) 40 mW is calculated if $100\ \Omega$ is incorrectly used for the resistance of R_M instead of $200\ \Omega$.

(Choice C) 200 mW is calculated if the conversion from W to mW incorrectly uses a factor of 100 instead of a factor of 1000.

(Choice D) The electric power equation can also be written in terms of current and resistance: $P = I^2R$. Incorrectly using this equation with V instead of I will yield $P = 800\text{ W}$.

Educational objective:

The power dissipated by a resistor is released as heat, and is the product of voltage and current: $P = IV$. Ohm law ($V = IR$) can be used to determine alternate expressions for electric power in terms of resistance: $P = \frac{V^2}{R} = I^2 R$.

Time spent:0 sec**Subject:**
Physics**QID:**401067**System:**
Electrostatics and Circuits

Cable theory is a mathematical model used to calculate currents and voltages along and across axon membranes. The model treats small sections of the membrane as the resistor-capacitor circuit shown in Figure 1. An electric neuronal signal enters the circuit through the input node. A portion of the signal remains inside the axon and continues to the next section of membrane via Output 1, and the rest of the signal exits to the extracellular fluid via Output 2.

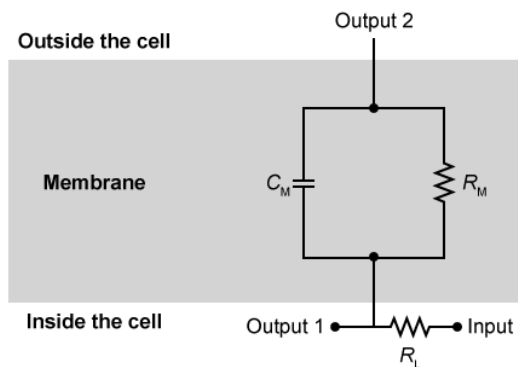


Figure 1 Circuit unit used to model axon membranes

Researchers studied cable theory by recreating the circuit in Figure 1 using a $2\ \mu\text{F}$ capacitor for C_M , a $200\ \Omega$ resistor for R_M , and a $1,000\ \Omega$ resistor for R_L . The researchers sent an electrical signal through the input and discovered that the capacitor could be modeled as a resistor whose resistance varied with the signal's frequency, as shown in Figure 2.

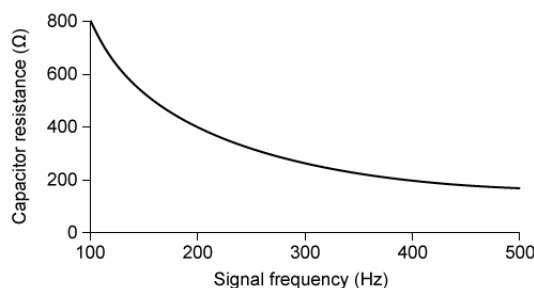


Figure 2 Capacitor equivalent resistance vs. signal frequency for C_M

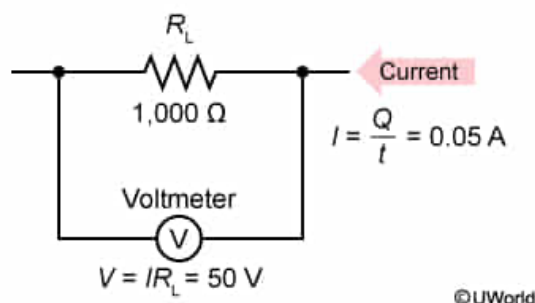
A voltmeter is placed across R_L . If a charge of $0.1\ \text{C}$ flows into the input node every $2\ \text{s}$, what will be the reading on the voltmeter?

- ☐ A. $0\ \text{V}$ [3%]
- ✓ ☒ B. $50\ \text{V}$ [74%]
- ☐ C. $100\ \text{V}$ [15%]
- ☐ D. $200\ \text{V}$ [6%]

Correct

74% answered correctly

Explanation:



Electric currents are formed by the motion of **electric charges** and are measured in **amperes (A)**. According to circuit sign conventions, current is positive in the direction of the flow of positive charges. If an electric charge Q passes through a point over a given time t , the current I is:

$$I = \frac{Q}{t}$$

Therefore, the current through R_L can be calculated as:

$$I = \frac{0.1\text{ C}}{2\text{ s}} = 0.05\text{ A}$$

When a current flows through a resistor, a voltage drop occurs across the resistor. Voltage is the difference in **electric potential** between two points, and it drives the movement of electric charges. A **voltmeter measures** the voltage between two points, and therefore measures the **voltage drop across R_L** when connected in parallel with R_L . The voltage drop across a resistor is calculated from **Ohm law**, which states that the voltage V across a resistor is the product of the current and its resistance R :

$$V = IR$$

Therefore, the reading on the voltmeter can be calculated from the product of I and R_L :

$$V = (0.05\text{ A})(1,000\ \Omega) = 50\text{ V}$$

(Choice A) Because the voltage drops across a resistor and voltmeters measure voltage differences, the reading on the voltmeter cannot be zero if there is a current flowing through the resistor.

(Choice C) If the electric charge of 0.1 C is incorrectly used as the current through the resistor, the voltage is incorrectly calculated as 100 V .

(Choice D) If current is incorrectly obtained by multiplying Q and t , voltage is incorrectly calculated as 200 V .

Educational objective:

Electric current is the rate of charge flow and is calculated as charge divided by time: $I = Q/t$. Voltage is the difference in electric potential between two points, and it drives the movement of electric charges. The voltage across a resistor can be calculated from Ohm law ($V = IR$).

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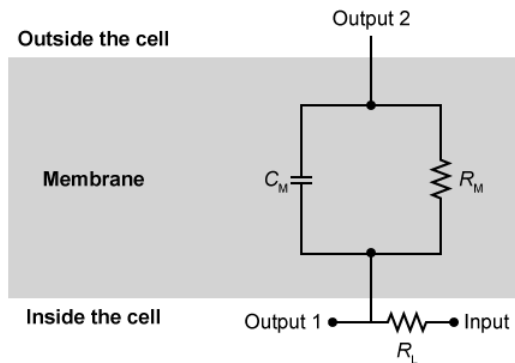


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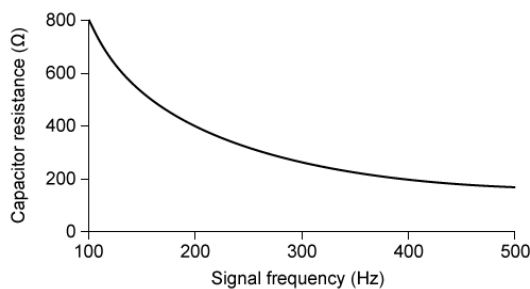
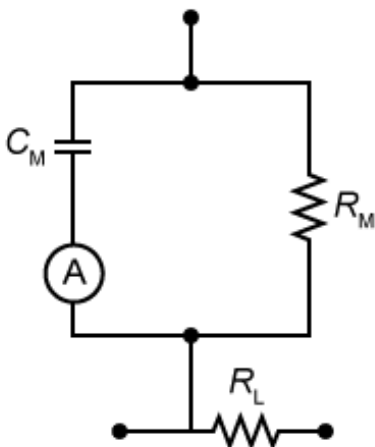


Figure 2 Capacitor equivalent resistance vs. signal frequency for C_M



An ammeter is added to the circuit as shown. The voltage across the capacitor C_M is kept constant while the frequency is increased from 200 Hz to 400 Hz. Through this change, the

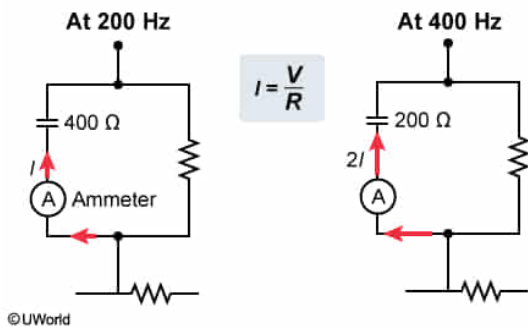
value shown by the ammeter:

- ☐ A. will decrease by 50%. [16%]
- ☐ B. will not change. [13%]
- ☐ C. will increase by 50%. [31%]
- ✓ ☒ D. will increase by 100%. [38%]

Correct

37% answered correctly

Explanation:



An **ammeter** in series with a circuit component **measures** the electric current flowing through it. In the given diagram, the ammeter is in series with the capacitor C_M , and therefore it measures the **current through C_M** . **Ohm's law** states that the voltage V across a resistor equals the product of its resistance R and the electric current I flowing through it ($V = IR$). Therefore,

$$I = \frac{V}{R}$$

In this question, V across C_M is held constant. Therefore, a **change in R** causes an **inverse change in I** . C_M is modeled as a resistor R_C with the resistance given in [Figure 2](#). At 200 Hz, the value of R_C is 400 Ω , and it decreases to 200 Ω at 400 Hz. As such, decreasing R_C to half its value doubles I :

$$I = \frac{V}{R/2} = 2\frac{V}{R}$$

Therefore, the current reading on the ammeter will double, or **increase by 100%**.

(Choice A) If 400 Hz is incorrectly interpreted as the initial frequency value and 200 Hz as the final value, the resistance would appear to double and the current appear to decrease by 50%.

(Choice B) If the ammeter is incorrectly assumed to measure voltage instead of current, the value shown by the ammeter would not appear to change because the voltage is kept constant.

(Choice C) The value of R_C decreases by 50%, but that does not cause a 50% increase in the current. Instead, the current increases by 100%.

Educational objective:

An ammeter in series with a circuit component measures the current through that component. Ohm law states that voltage equals the product of current and resistance: $V = IR$. Because current is inversely proportional to resistance if the voltage is fixed, reducing the resistance by half will double the current.

Time spent:0 sec**Subject:**
Physics**QID:**401069**System:**
Electrostatics and Circuits

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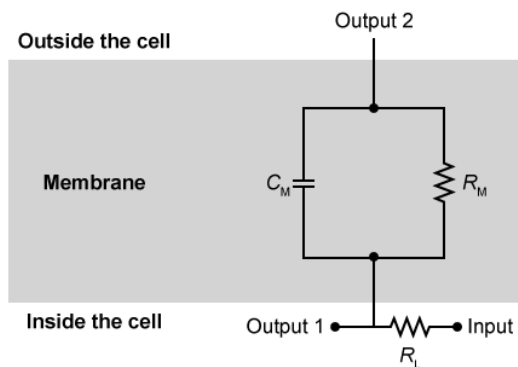


Figure 1 Circuit unit used to model axon membranes

Researchers studied cable theory by recreating the circuit in Figure 1 using a $2\ \mu\text{F}$ capacitor for C_M , a $200\ \Omega$ resistor for R_M , and a $1,000\ \Omega$ resistor for R_L . The researchers sent an electrical signal through the input and discovered that the capacitor could be modeled as a resistor whose resistance varied with the signal's frequency, as shown in Figure 2.

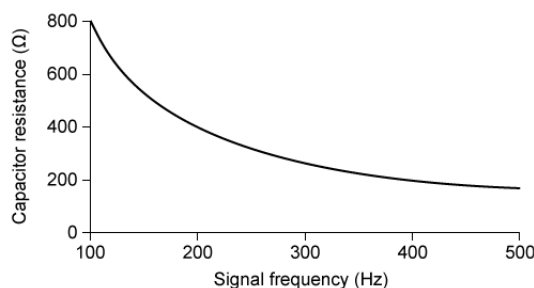


Figure 2 Capacitor equivalent resistance vs. signal frequency for C_M

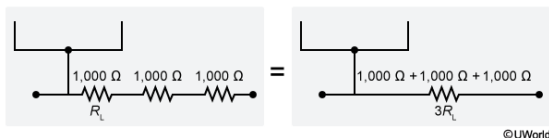
Adding two $1,000\ \Omega$ resistors in series with R_L would be equivalent to:

- ☐ A. multiplying R_L by $1/3$ [6%]
- ☐ B. multiplying R_L by $2/3$ [3%]
- ☐ C. multiplying R_L by 2. [16%]
- ☒ D. multiplying R_L by 3. [72%]

Correct

73% answered correctly

Explanation:



Electrical **components in series** are **connected end to end**, leaving only one path for the current to flow through the circuit. The properties of resistors in series are:

- The current through each resistor is the same.
- The voltage drop across each resistor is independent from that of the other resistors.
- The total voltage drop across all the resistors is equal to the sum of the individual voltage drops of the individual resistors.

For resistors in series, the **equivalent resistance** is the sum of the individual resistances:

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

The value of the resistor R_L is $1,000 \, \Omega$. If two $1,000 \, \Omega$ resistors are added in series with R_L , the equivalent resistance is the sum of the three individual resistances:

$$R_{eq} = 1,000 \, \Omega + 1,000 \, \Omega + 1,000 \, \Omega = 3(1,000) \, \Omega$$

Therefore, adding two $1,000 \, \Omega$ resistors in series with R_L (also $1,000 \, \Omega$) is equivalent to **multiplying R_L by 3**.

(Choice A) The equivalent resistance of a **parallel circuit of resistors** is calculated from:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Using the above equation, the equivalent resistance for a parallel circuit is $R_{eq} = \frac{1,000 \, \Omega}{3}$, which is the equivalent of multiplying R_L by $\frac{1}{3}$.

(Choice B) Because the equivalent resistance of a series circuit of resistors is the sum of the individual resistances, the equivalent resistance cannot decrease.

(Choice C) The equivalent resistance of the two $1,000 \, \Omega$ resistors added in series to the circuit is equivalent to multiplying R_L by 2, but the circuit already contains the resistor R_L . Therefore, the modified circuit contains three $1,000 \, \Omega$ resistors in series, which is equivalent to multiplying R_L by 3.

Educational objective:

Circuit components in series are connected end to end and share the same current. The equivalent resistance of resistors in series is the sum of the individual resistances.

Time spent: 0 sec

Subject:
Physics

QID: 401070

System:
Electrostatics and Circuits